Table 1: Prominent tools for testing Web applications

Tool	Function and Key Operations	Official URL
Apache JMeter	Load testing Performance testing	*https://jmeter.apache.org/*
Locust	Load testing Performance testing	*https://locust.io/*
Grinder	Load testing	*http://grinder.sourceforge.net/*
Multi-Mechanize	Load testing Performance testing scalability testing	*https://multi-mechanize.readthedocs.io/en/latest/*
Selenium	Automation of web browsers, Regression automation exploratory testing	*https://www.seleniumhq.org/*
Capybara	Simulation of user behavior	*https://github.com/teamcapybara/capybara*
OpenSTA	Stress testing web load testing,	*http://opensta.org/*
Pylot	Load testing benchmarking capacity planning system tuning	*https://code.google.com/archive/p/pylt/downloads*
WebLoad	Load testing, Response validation testing	*https://www.radview.com/webload-download/*
Webrat	Acceptance testing browser simulation	*https://github.com/brynary/webrat*
Appium	Testing of hybrid, native and mobile web apps	*http://appium.io/*
Cucumber	Acceptance testing	*https://cucumber.io/z*
Watir	Automation testing	*http://watir.com/*
Canoo Web Test	Automation testing	*http://webtest.canoo.com/*
SoapUI	SOAP testing REST testing	*https://www.soapui.org/*
Selendroid	Automation testing for mobile apps	*http://selendroid.io/*
Gatling	Performance testing Load testing	*https://gatling.io/*
Tsung	Stress testing Distributed load testing	*http://tsung.erlang-projects.org/*
FItNesse	Acceptance testing	*http://fitnesse.org/*
TestNG	Server testing performance testing Data driven testing	*https://testng.org*

on multiple parameters (including the number of requests, content size, failure attempts, etc) is presented.

Locust has plain code without any complex settings. The easiest way to install it is from PyPI, using Pip, as shown below:

```
> pip install locustio
```

The code of Python *locustfile.py* can be directly executed on a command prompt to test the Web application.

The key features of Locust are:

- Free and open source under Apache
- Command line interface (CLI) based
- Evaluates both standalone or live websites
- Conducts performance testing for Web servers
- Specifically for HTTP Web Server
- Load testing and benchmarking
- Platform independent

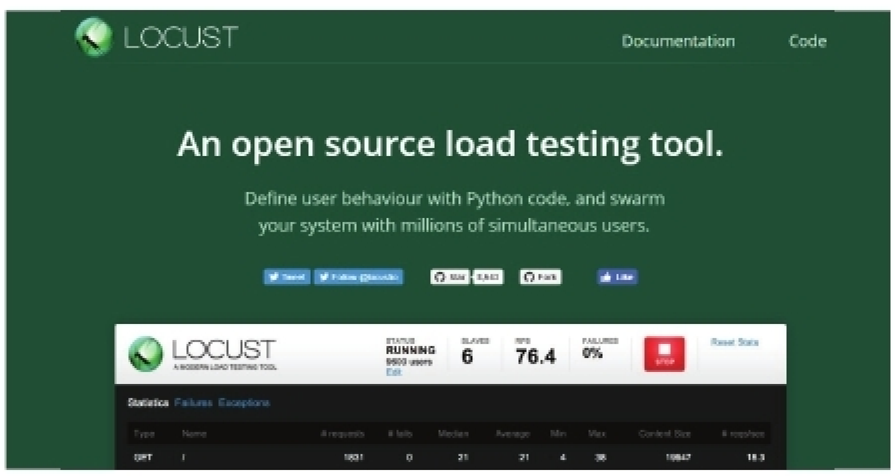

Figure 1: Official website of Locust

Apache JMeter

URL: *https://jmeter.apache.org*

Apache JMeter is another free and open source tool that is used for load testing, performance testing and to gauge the functional behaviour of Web applications. It was developed for the audit of Web applications, but nowadays it is also used for core network functions and the deep evaluation of protocols.

Apache JMeter is able to test and present a deep audit of the following:

- File Transfer Protocol (FTP)
- Directory services
- Lightweight Directory Access Protocol (LDAP)
- Database connections
- TCP

locustfile.py

```
from locust import HttpLocust, TaskSet, task

class WebsiteTasks(TaskSet):
    def on_start(self):
        self.client.post("/login", {
            "username": "test_user",
            "password": ""
        })

    @task
    def index(self):
        self.client.get("/")

    @task
    def about(self):
        self.client.get("/about/")

class WebsiteUser(HttpLocust):
    task_set = WebsiteTasks
    min_wait = 5000
    max_wait = 15000
```

```
$ locust -f locustfile.py
```

Figure 2: The view of *locustfile.py* and the instruction to run